KIA TASMAN X-PRO FIT-OUT · v0.14 · 2026-10-20

Vehicle measurement sheet

Every dimension the 3D model still guesses: 104 items in nine areas. Fill these in, send back a photo of the pages (or the CSV), and the model gets updated. Then the cutting lists can be trusted. Each item shows **what the model assumes now**. A big difference from that is exactly what we need to catch. The **Stage** column shows the first build stage that depends on it (see docs/build-stages.md), and the 📷 box is for ticking off a photo.

Before you start

- Park on level ground, normal tyre pressures, unladen, and note the fuel level.
- Everything is in **mm**, measured to the **liner surface** (what the furniture will touch) unless it says otherwise.
- Driver side = right (D), passenger side = left (P).
- Run masking tape along the tub centreline (half the wall-to-wall width, marked at the front and at the tailgate). That's Y = 0.
- "Front / mid / rear" in the tub means X ≈ 100, 750 and 1400 unless an item says otherwise.
- Photograph each item with the tape in shot and its ID written on masking tape.
- Measure twice. If two readings differ by more than 2 mm, write both.

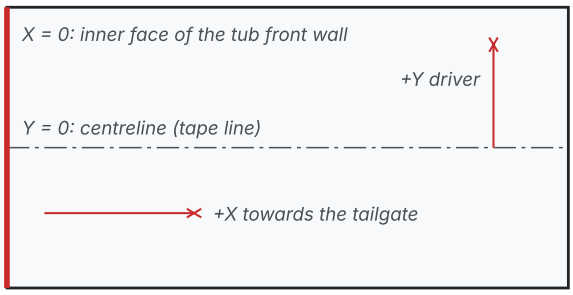
Tools

- 5 m tape and a 1 m steel rule
- 300 mm steel rule and digital calipers
- Straight edge ≥ 1.8 m (an aluminium level is ideal) and a spirit level
- Digital angle finder (a phone app works)
- Plumb bob or a laser line, string line
- Masking tape, marker, a radius gauge or a few coins

Priority

- **Stages 1 and 3** (tub, arches, fittings, chassis): needed before cutting anything for the battery bay, cabinets and slides.
- **Stage 6** (lid, bars, tent): under-tent storage.
- **Stages 2 and 7** (roof, windscreen): platform and visor.

D0 · Reference lines (datums) used on every sheet



plan view, front of the vehicle on the left

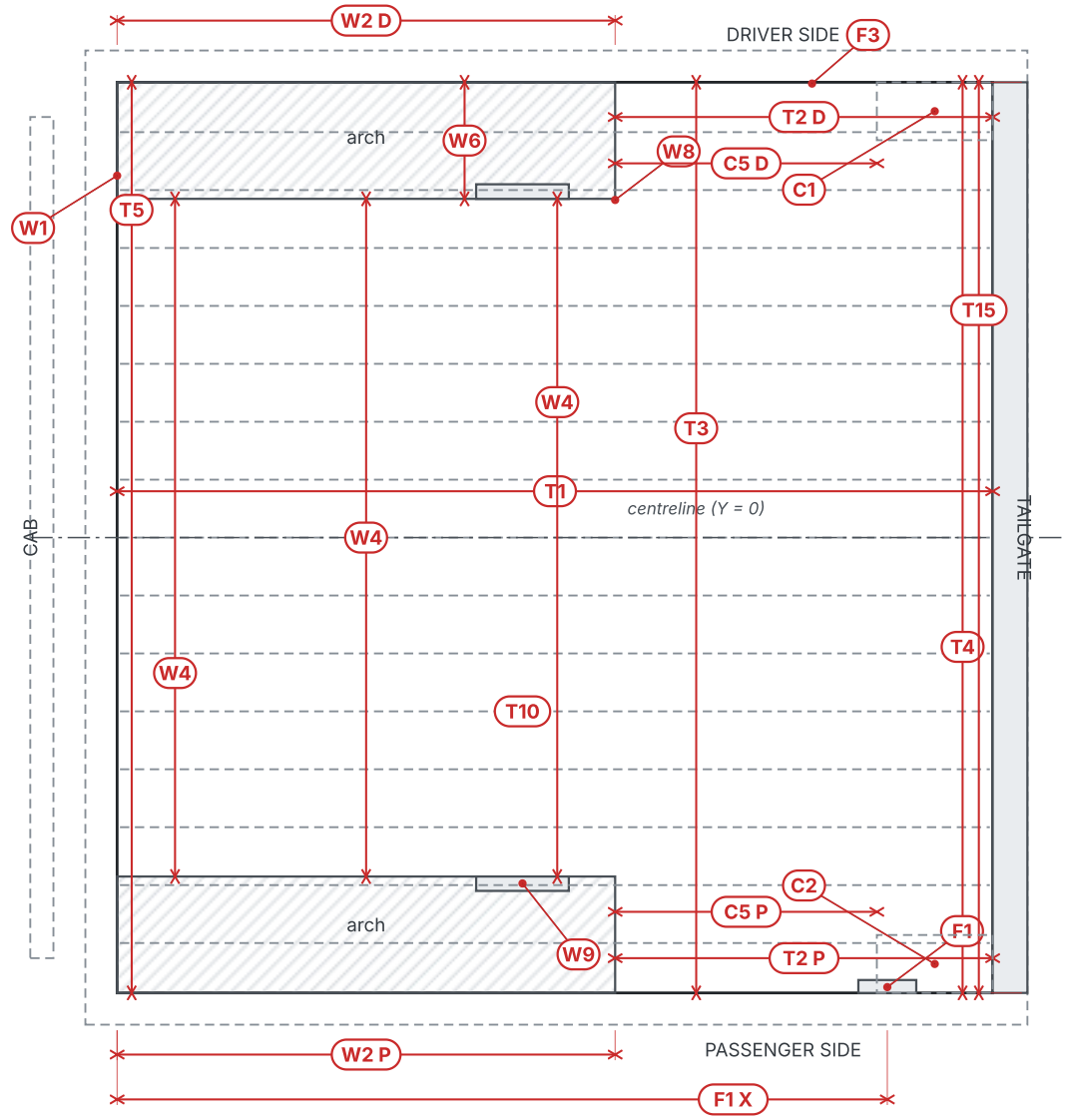


side view

Generated from cad/params.py by tools/measure/measure_sheet.py. The diagrams are schematic, drawn from the model's current guesses, so they may not look exactly like your ute. Measure what's really there.

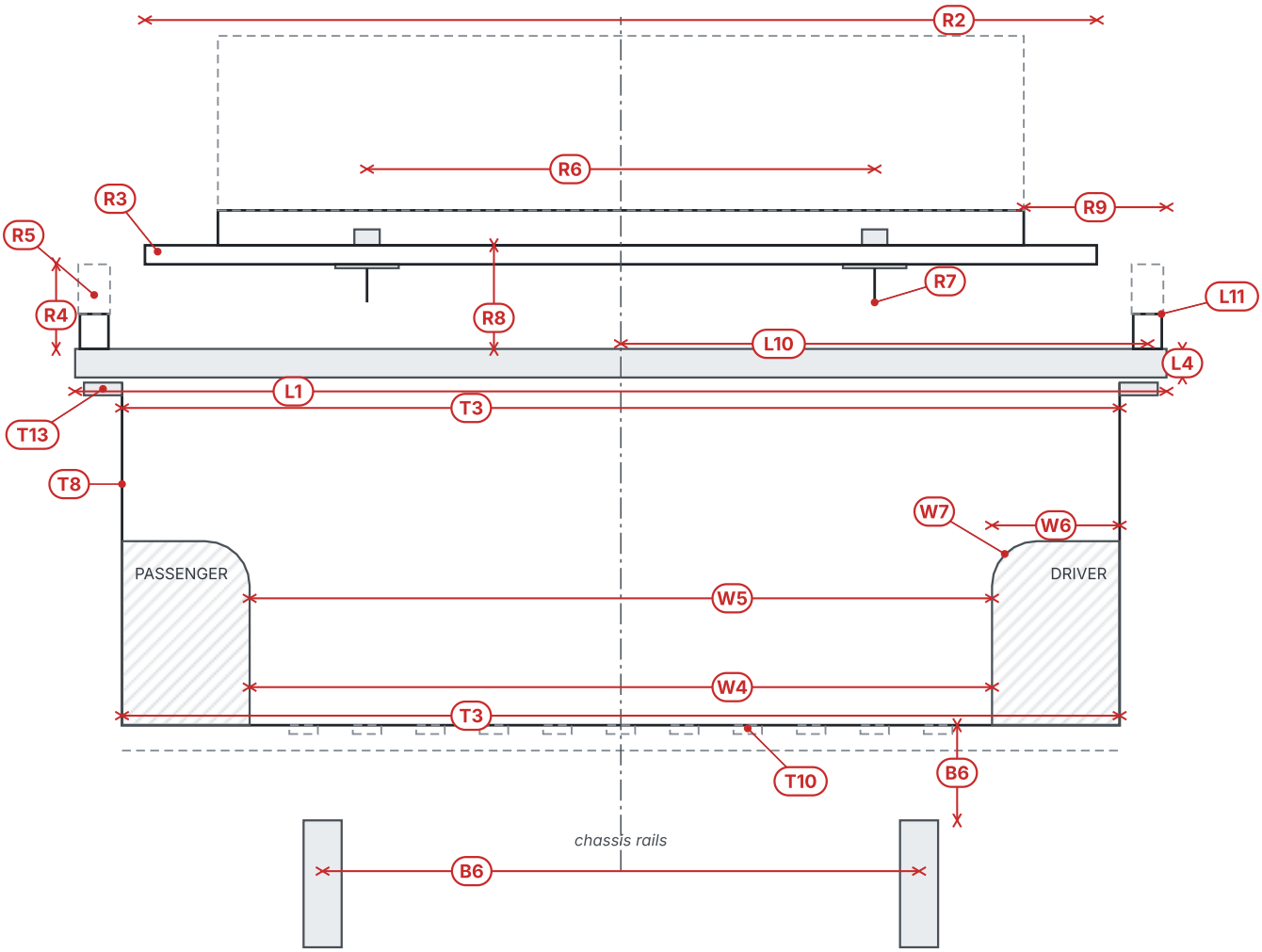
1 · Tub: floor, walls and tailgate

D1 · Tub, plan view (from above)



X from the tub front wall (liner face) · Y from the centreline · all in mm · ribs shown at the assumed 100 mm pitch

D3 · Cross-section looking forward from the tailgate (driver side on the right)



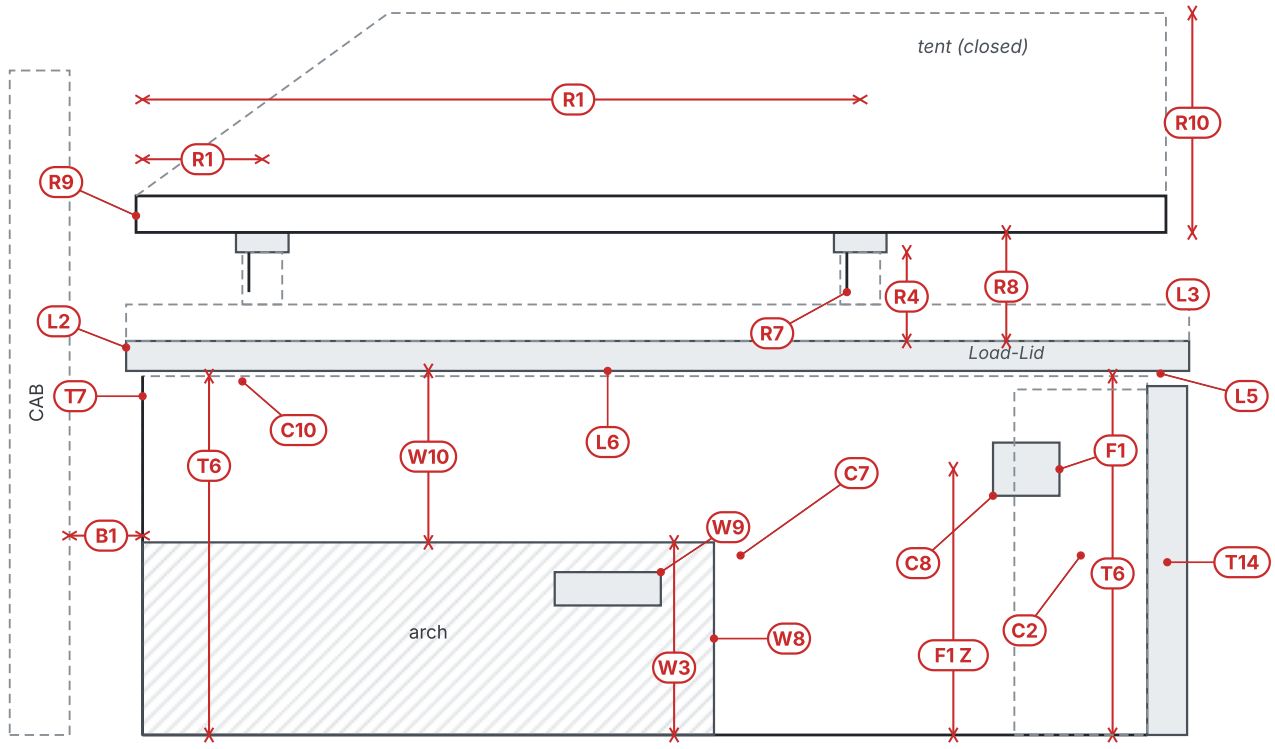
Diagrams: D1, D3

ID	Measure	Model now	Your values (mm unless noted)	Stage	📷
T1	**Floor length on the centreline** Front wall liner face → inside face of the closed tailgate, at rib-top height	1512 TUB_L	mm	1 3	☐
T2	**Floor length beside the arches** Rear end of each arch → tailgate inside face, 50 mm in from the side wall	652 TL – ARCH_L	Driver / Passenger	3	☐
T3	**Width wall to wall behind the arches** Liner face to liner face at X ≈ 1000: at floor, at 270 high, and just below the rail	1572 TUB_W	Floor / 270 high / Rail	1 3	☐
T4	**Width wall to wall at the tailgate** Same, at X ≈ 1460	1572 TUB_W	Floor / Rail	3	☐
T5	**Width above the arches at the front** Wall to wall at X ≈ 50, just above arch height and at rail height	1572 TUB_W	Above arch / Rail	1	☐
T6	**Tub depth** Rib top → top of the tub rail (inside edge), at each corner	540 TUB_D	Front D / Front P / Rear D / Rear P	1 3	☐

T7	**Front wall lean** Level held vertical against the front wall: gap at rail height (+ = wall leans back)	0	mm	1	☐
T8	**Side wall lean** Same on each side wall behind the arch (+ = tub wider at the top)	0	Driver / Passenger	3	☐
T9	**Corner radii** Floor-to-wall radius, and the radius in the front corners (vertical)	–	Floor/wall / Front corners	1 3	☐
T10	**Floor ribs (detail a)** Rib height above the valleys, rib top width, pitch, how many, Y of the middle rib, X where they start and end	14 / 45 / 100 `RIB_H, RIB_W, RIB_PITCH`	Height / Top width / Pitch / Count Mid rib Y / Start X / End X	1 3	☐
T11	**Liner thickness (detail f)** Calipers on an exposed liner edge (rail lip or tailgate gap)	8 `LINER_ALLOW`	mm	3	☐
T12	**Liner type** Drop-in or spray-in? How is it fixed? Can it come out to bolt through the floor?	drop-in	.....	1	☐
T13	**Tub rail cap (detail f)** Width of the rail's flat top, how far its inside lip overhangs the wall, lip depth	–	Cap width / Lip overhang / Lip depth	3	☐
T14	**Tailgate inner panel** Thickness inside face → outer skin at mid-height; note any handles or bulges facing the tub	60 `TAILGATE_T`	Thickness / Bulges?	3	☐
T15	**Tailgate opening** Clear width between the latch pillars at the floor and at the top; height of the closed tailgate top above the floor	– / 525 `TAILGATE_H`	Width floor / Width top / Height	3	☐
T16	**Tailgate open** Tailgate down: its top surface vs the tub floor (+ above / – below), its length from the hinge, the cable stay rating	–	Height diff / Length / Rating kg	3	☐
T17	**Holes in the floor** X, Y of any drain holes or existing bolt holes (lift the liner if it's drop-in)	–	.....	1	☐

2 · Wheel arches

D2 · Tub, side section (looking at the passenger wall from the centre)



Diagrams: D1, D2, D3

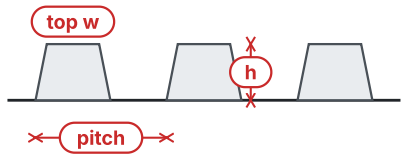
ID	Measure	Model now	Your values (mm unless noted)	Stage	📷
W1	**Arch start** Gap from the front wall to the front of each arch at the floor and at the top (0 if they meet)	0	D floor ☐ D top ☐ P floor ☐ P top ☐	1	☐
W2	**Arch length** Front → rear end of each arch along its inner face, at the floor and at the top	860 ARCH_L	D floor ☐ D top ☐ P floor ☐ P top ☐	1 3	☐
W3	**Arch height** Rib top → top of each arch at the front, middle and rear of its flat top	290 ARCH_H	D front ☐ D mid ☐ D rear ☐ P front ☐ P mid ☐ P rear ☐	1 3	☐
W4	**Width between the arches (floor)** Arch face to arch face just above the floor at X ≈ 100, 430, 760	1170 ARCH_GAP	Front ☐ Mid ☐ Rear ☐	1 3	☐
W5	**Width between the arches (top)** Same three positions, just below the top radius	1170 ARCH_GAP	Front ☐ Mid ☐ Rear ☐	1 3	☐
W6	**Arch intrusion** Side wall face → arch inner face at mid-length	201 ARCH_IN	Driver ☐ Passenger ☐	3	☐

W7	**Arch top shape** Flat width of the top (wall → start of the radius) and the radius on its inboard edge	– / 70 ARCH_R	D flat D radius P flat P radius	3	☐
W8	**Arch rear end** Vertical, sloped or curved? How far the top's rear edge sits forward of the floor's rear edge	vertical	Driver Passenger	3	☐
W9	**Arch well (detail e)** The recess in each arch face: X of its front and rear edges, Z of its bottom and top, depth	620–780 / 195–245 / 35 ARCH_POCKET	D x0 D x1 D z0 D z1 D depth P x0 P x1 P z0 P z1 P depth	3	☐
W10	**Space above the arch** Arch top → underside of the closed Load-Lid (lowest part of its frame), front and rear of each arch	258 UNDER_LID_Z − ARCH_H	D front D rear P front P rear	3	☐

3 · Factory fittings in the tub

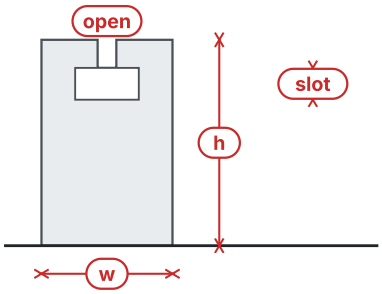
D7 · Details (not to scale)

a · Floor ribs (T10)



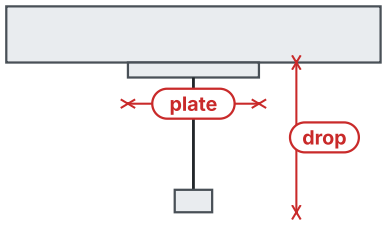
rib height · top width · pitch · count

b · Destroyer rail + T-slot (L10, L11)



height · width · slot opening · slot depth · inner width

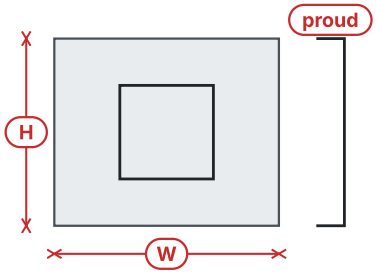
c · Tent bolt at each bar (R7)



Vortex bar · clamp plate · bolt + nut

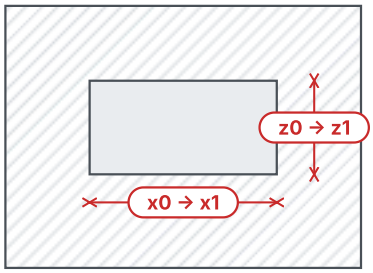
measure bar underside → lowest point

d · Factory 240 V outlet (F1, F2)



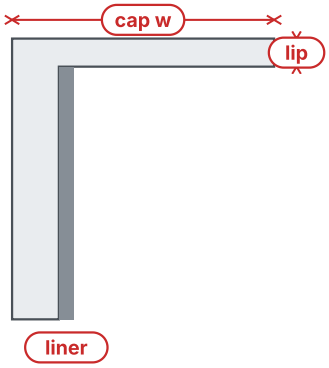
housing W × H · how far it stands proud

e · Arch well (W9)



X of front/rear edge from the front wall,
Z of bottom/top above the floor, depth, radius

f · Liner and rail cap (T11, T13)



cap width · lip overhang · liner thickness

Diagrams: D1, D2, D7

ID	Measure	Model now	Your values (mm unless noted)	Stage	📷
F1	**Factory 240 V outlet (detail d)** Centre X from the front wall, centre Z above the floor, housing width × height, how far it stands proud	1330 / 400 FACTORY_OUTLET	X ☐ Z ☐ W ☐ H ☐ Proud ☐	1	☐
F2	**Behind the outlet** With the outlet removed: hole size in the steel, cavity depth behind (model: 49 mm between liner and skin), and the way out at the cavity bottom for a 50 mm and a 25 mm conduit (routes R14 / R15, sheet E12)	– / 49	Hole W ☐ Hole H ☐ Cavity ☐ Exit ☐	1	☐
F3	**Tie-down points** Every tie-down or rail: wall, X, Z, type and how far it stands proud	none on the arches	.....	1 3	☐
F4	**Other protrusions** Bolt heads, cleats, liner bosses, light: position and how far they stand proud	–	.....	3	☐

3b · Rear corners: tail-light housings (added 2026-10-25)

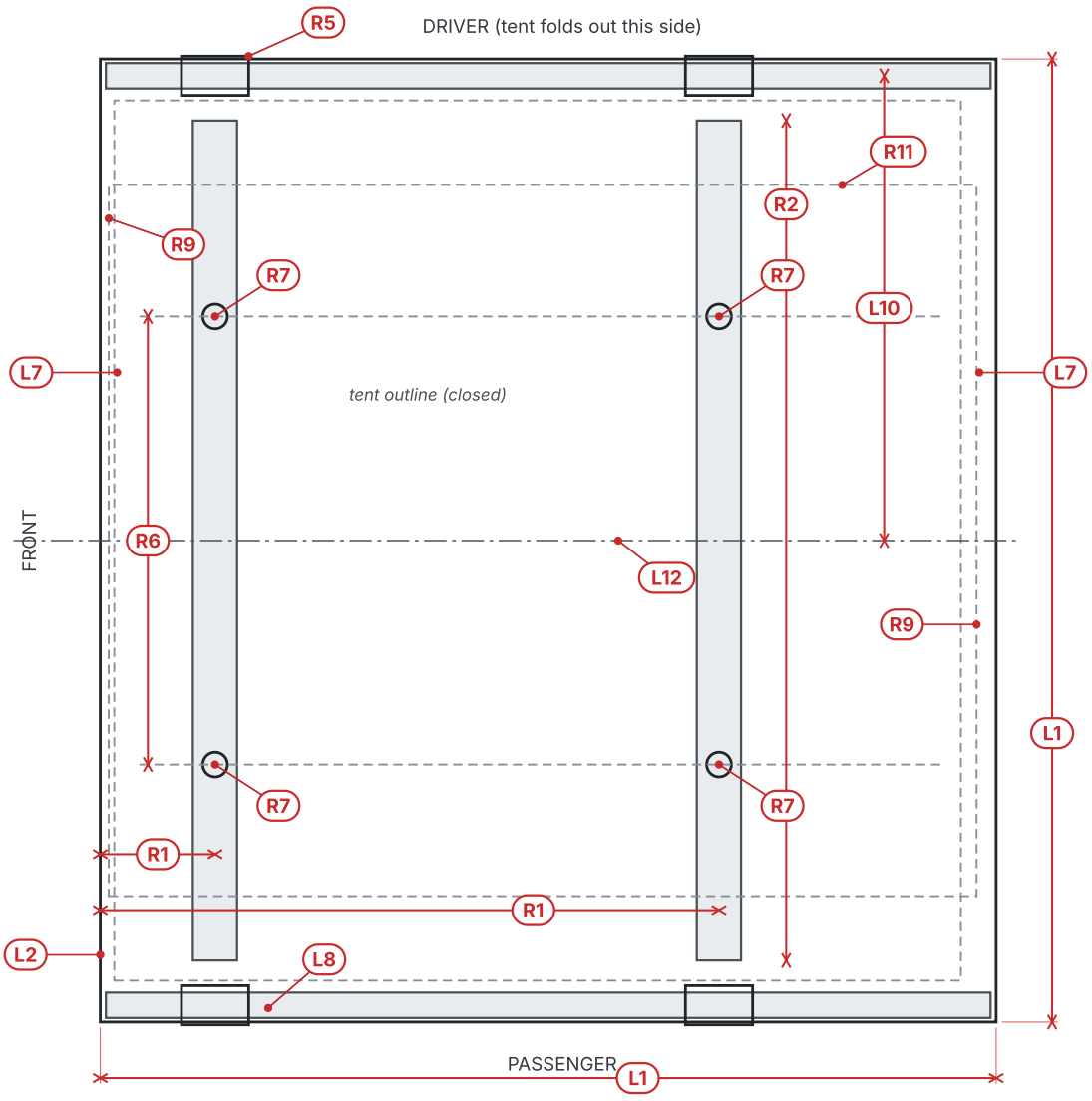
Diagrams: D1, D2

ID	Measure	Model now	Your values (mm unless noted)	Stage	📷
C1	**Tail-light housing, driver** The bulge in the side wall at the rear corner. X where its front face starts (from the front wall) and X where it meets the tailgate seal; how far it stands in from the side wall at the floor, 270 high and just below the rail; Z of its bottom and top	none `TAIL_BULGE_L, TAIL_BULGE_IN`	X start ☐ X end ☐ In floor ☐ In 270 ☐ In rail ☐ Z bottom ☐ Z top ☐	1 3	☐
C2	**Tail-light housing, passenger** Same as C1 on the passenger (outlet) side	none `TAIL_BULGE_L, TAIL_BULGE_IN`	X start ☐ X end ☐ In floor ☐ In 270 ☐ In rail ☐ Z bottom ☐ Z top ☐	1 3	☐
C3	**Housing shape** Flat or curved face? Square, sloped or rounded front edge (sketch it)? Any lip where it meets the wall? Same both sides?	–	....	3	☐
C4	**Access plugs** Each push-rivet plug or cap on the housings: X, Z, diameter, what's behind it (lamp bolts, globe, wiring) and how much hand room it needs. Anything built here must leave these reachable	–	D X ☐ D Z ☐ D dia ☐ P X ☐ P Z ☐ P dia ☐ Hand room ☐	3	☐
C5	**Clear length behind the arch** Rear end of the arch → front face of the tail-light housing, 50 mm in from the side wall, at the floor, 270 high and just below the rail	652 `TL − ARCH_L`	D floor ☐ D 270 ☐ D rail ☐ P floor ☐ P 270 ☐ P rail ☐	3	☐
C6	**Clear width beside the housing** Inner face of the housing → centreline of the tub at 270 high (shows how much is left outboard of the slides)	786 `TUB_W / 2`	Driver ☐ Passenger ☐	3	☐
C7	**Rib behind the arch** The vertical pressed rib between the arch and the outlet recess: X of its centre, width, how far it stands proud, full height or not	–	D X ☐ D width ☐ D proud ☐ P X ☐ P width ☐ P proud ☐	3	☐
C8	**Outlet recess** The recessed panel the factory outlet sits in: X of its front and rear edges, Z of its bottom and top, depth behind the wall face (photos show it lower and further forward than the model)	1255–1405 `FACTORY_OUTLET`	X front ☐ X rear ☐ Z bottom ☐ Z top ☐ Depth ☐	1	☐
C9	**Arch top profile** Photos show a stepped arch top: X of each step and the height before and after it, each side; how far the rear end slopes	flat at 290 `ARCH_H`	....	1 3	☐

C10	**Lid hardware hanging into the tub** Every clamp, star knob, bracket and gas strut below the rail: side, X, how far below the rail top, how far in from the wall (photos: knobs and brackets along both rails, a strut at the front corner)	nothing below 548 `UNDER_LID_Z`		1 3	☐

4 · Load-Lid and Destroyer rails

D4 · Load-Lid, bars and tent, plan view (from above)



Diagrams: D2, D3, D4

ID	Measure	Model now	Your values (mm unless noted)	Stage	📷
L1	**Lid size** Overall length (along) and width (across) including edge trims	1600 × 1720 `LID_L, LID_W`	Length ☐ Width ☐	6	☐
L2	**Lid front edge** Lid front edge relative to the tub front wall face (− = ahead of the wall)	-25 `LID_X0`	mm ☐	6	☐
L3	**Lid rear edge** Lid rear edge relative to the closed tailgate's outer face (+ = overhangs)	–	mm ☐	6	☐
L4	**Lid thickness** Top skin → lowest part of its perimeter frame	45 `LID_T`	mm ☐	3 6	☐

L5	**Gap rail → lid** Tub rail top → lid underside at the four corners	8 `LID_CLEAR`	FD ☐ FP ☐ RD ☐ RP ☐	3	☐
L6	**Lid frame inside the tub** Anything below the tub rail level (cross members, brackets, seals): what, X position, how far down	nothing `UNDER_LID_Z`	Max drop ☐	3	☐
L7	**Hinges and lock** Hinge type and positions; lock/latch positions and how far they reach into the tub	–	.....	3	☐
L8	**Gas struts** Both ends of each strut: on the lid / on the body, X, Y, Z. Inside the tub?	not in the tub	.....	3	☐
L9	**Lid clamps** How the lid bolts to the tub; clamp positions and how far they stick inside the tub walls	–	.....	3	☐
L10	**Destroyer rails (detail b)** Height above the lid top, width, length, X of the front and rear ends, Y of the rail centre from the centreline	55 / 45 / – / 830 `LID_RAIL_H, LID_RAIL_W, LID_RAIL_Y`	Height ☐ Width ☐ Length ☐ Front X ☐ Rear X ☐ Y ☐	6	☐
L11	**Rail T-slot (detail b)** Slot opening, slot depth, inner width, lip thickness; T-bolt size used	–	Opening ☐ Depth ☐ Inner w ☐ Lip ☐ Bolt ☐	6	☐
L12	**Lid top surface** Chequer pattern height; flat or crowned (straight edge across: gap at the centre)	flat	Pattern ☐ Crown ☐	6	☐
L13	**Lid underside over the batteries (v0.14)** Tub floor → lid underside at the driver front corner (X ≈ 100 and 250, Y ≈ 200 and 500): the battery top foam assumes 9 mm above the handle lugs	548 `UNDER_LID_Z`	X100 Y200 ☐ X100 Y500 ☐ X250 Y200 ☐ X250 Y500 ☐	1	☐
L14	**Space for the water-can jacket (v0.14)** Lid top → tent base underside over the water can, and lid top → the tent's under-rail; is a 30 mm jacket top possible?	163.4 `TENT_Z0 − LID_TOP_Z`	To base ☐ To rail ☐	6	☐

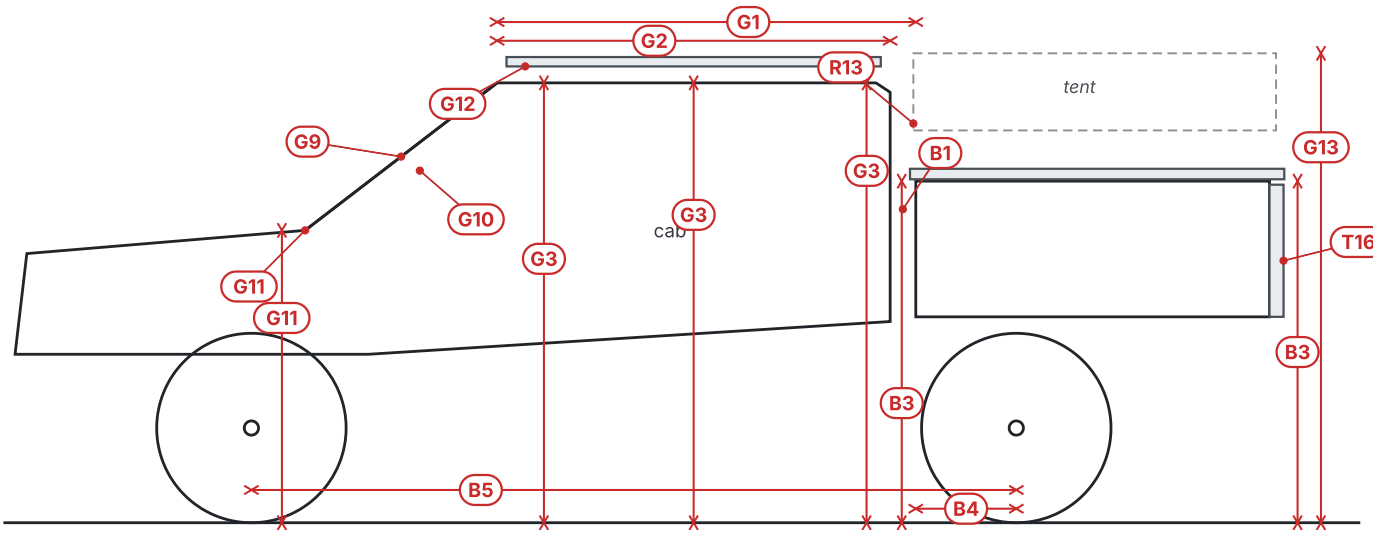
5 · Rack bars and tent

Diagrams: D2, D3, D4, D7

ID	Measure	Model now	Your values (mm unless noted)	Stage	📷
R1	**Bar positions** Centre line of each Vortex bar from the lid front edge (or from the tub front wall, say which)	180 / 1080 (from tub wall) `RACK_BAR_X`	Front bar ☐ Rear bar ☐ Datum ☐	6	☐
R2	**Bar length and overhang** Overall length; overhang past each leg	1500 `RACK_BAR_L`	Length ☐ D overhang ☐ P overhang ☐	6	☐
R3	**Bar section** Width × height of the bar; top slot width	79 × 30 `RACK_BAR`	W ☐ H ☐ Slot ☐	6	☐
R4	**Bar height** Lid top → underside of each bar beside each leg	133.4 `TENT_GAP − bar`	Front D ☐ Front P ☐ Rear D ☐ Rear P ☐	6	☐
R5	**Legs / brackets** Footprint on the rail (along × across), leg height, part number	–	Footprint ☐ Height ☐ Part no. ☐	6	☐
R6	**Tent under-rails** Y of each under-rail centre from the centreline; X of the rail ends	±400 `TENT_UNDER_RAILS_CC`	D Y ☐ P Y ☐ Front X ☐ Rear X ☐	6	☐
R7	**Tent bolt tails (detail c)** At each of the 4 crossings: bar underside → lowest point of bolt, nut or plate; plate size; bolt size	60 `TENT_BOLT_DROP`	Front D ☐ Front P ☐ Rear D ☐ Rear P ☐ Plate ☐ Bolt ☐	6	☐
R8	**Gap lid → tent** Lid top → lowest part of the tent base at the front, middle and rear, both sides	163.4 `TENT_GAP`	Front D ☐ Front P ☐ Mid D ☐ Mid P ☐ Rear D ☐ Rear P ☐	6	☐
R9	**Tent position** Tent front and rear edges from the lid front edge; overhang past the lid edge each side	-10 `TENT_X0`	Front ☐ Rear ☐ D overhang ☐ P overhang ☐	6	☐
R10	**Tent shape** Closed height (base → top of the roof bars), length of the front slope, top-bar spacing and length	330 / 380 / 1040 / 890 `TENT_H, TENT_SLOPE_L, TENT_BAR_CC, TENT_BAR_L`	Height ☐ Slope ☐ Bar c/c ☐ Bar length ☐	5	☐
R11	**Ladder bracket** Where it is and how far it sticks out or down	–	.....	6	☐
R12	**Tent open** How far the open tent reaches past the driver side; floor height when open	–	Reach ☐ Floor height ☐	-	☐
R13	**Tent to cab** Tent front face → cab rear at tent-base height	–	mm ☐	4	☐

6 · Body, cab gap and chassis

D5 · Vehicle side view (passenger side): heights from the ground, positions from the tub front wall



Park on level ground, normal tyre pressures, unladen. Use a plumb line or level to transfer points to the ground.

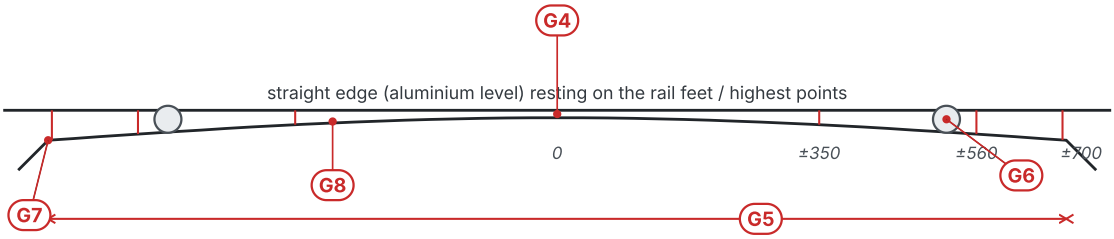
Diagrams: D5, D3

ID	Measure	Model now	Your values (mm unless noted)	Stage	📷
B1	**Cab–tub gap** Cab back wall → tub front wall at the top, middle and bottom	55 CAB_REAR_X	Top ____ Mid ____ Bottom ____	1 4	☐
B2	**In the gap** Body mounts, hoses, lines, wiring. Free space near the passenger top corner for a ~150 × 150 × 90 junction box?	–	Free W ____ Free H ____ Free D ____	4	☐
B3	**Tub height** Ground → top of the tub rail at the tailgate and at the front corner (the floor is this minus T6)	1460 FLOOR_ABOVE_GROUND	Rear ____ Front ____	1	☐
B4	**Rear wheel centre** Plumb from the tub front edge and from the wheel centre; distance along the ground	430 REAR_AXLE_X	Driver ____ Passenger ____	1	☐
B5	**Wheelbase check** Front → rear wheel centres	3270 WHEELBASE	Driver ____ Passenger ____	-	☐
B6	**Chassis rails** Centre Y of each rail; top of the rail below the tub floor; rail width × height	±470 / -150 CHASSIS_RAIL_Y, CHASSIS_Z	D Y ____ P Y ____ Top Z ____ W ____ H ____	1	☐
B7	**Cable route** Outlet → passenger chassis rail → cab gap: obstacles, existing clips and holes	–	....	1	☐

B8	**Tub underside** Cross-member positions (X) under the tub floor, in case anything bolts through	–		1	☐

7 · Roof, windscreen and platform

D6 · Roof crown: straight edge across the roof (repeat at the front, middle and rear)



Measure the gap from the straight edge down to the roof skin at each mark, both sides. Rear view; driver side on the right.

Diagrams: D5, D6

ID	Measure	Model now	Your values (mm unless noted)	Stage	📷
G1	**Header to tub** Horizontal: top of the windscreen (header seal) → tub front wall (plumb line from the header)	1790 CAB_FRONT_X	mm	2 7	☐
G2	**Roof length** Header → rear edge of the roof skin	1680	mm	2	☐
G3	**Roof height** Ground → roof skin on the centreline, 200 mm behind the header, middle, 100 mm from the rear	1880 ROOF_Z	Front / Mid / Rear	2 7	☐
G4	**Roof crown (D6)** Straight edge across the roof: gap to the skin at each mark. Rows: front / mid / rear	flat	P 700 / P 560 / P 350 / 0 / D 350 / D 560 / D 700 Front Mid Rear	2 7	☐
G5	**Roof width** Skin width between the drip channels at the front, middle, rear	–	Front / Mid / Rear	2	☐
G6	**Roof mounting points** How many; X (from the header) and Y of each; thread; cover type	–	.....	2	☐
G7	**Roof rails** Height, foot positions; do they come off for the Wedgetail kit?	–	.....	2	☐
G8	**Roof items** Shark-fin antenna position and height; anything else on the roof	–	.....	2 7	☐

G9	**Windscreen angle** Angle finder flat on the glass at the centre, from horizontal	–	degrees	7	☐
G10	**Windscreen size** Length along the glass (header → bottom edge, centre) and width at the top	–	Length / Width	7	☐
G11	**Bonnet rear edge** Height above ground and horizontal distance from the header at the centre; top of the parked wipers	–	Height / From header / Wipers	7	☐
G12	**Platform (once fitted)** Platform underside above the roof skin at the front and rear; front edge vs the header	70 PLAT_Z	Front / Rear / Front edge	2 7	☐
G13	**Overall height** Ground → top of the closed tent; ground → top of the platform	2006	Tent / Platform	-	☐
G14	**Headlining and roof cables** How far the headlining drops to reach the roof points; path for a cable roof → cab (which pillar?)	–		2	☐

8 · Engine bay and cabin

ID	Measure	Model now	Your values (mm unless noted)	Stage	📷
E1	**Start battery** Type and size; terminal type; room for a MIDI fuse holder within 300 mm of the + terminal	–		1	☐
E2	**Fuse box** Location; fuse type (mini / micro2) for the add-a-fuse; an ignition-switched circuit to tap	–		1	☐
E3	**Alternator** Rated current on its label (amps)	unknown	Amps	1	☐
E4	**Engine bay → chassis** Path for the 25 mm² twin from the start battery to the passenger chassis rail (R21); exhaust distance along the rail (R17–R19)	–		1	☐
E5	**Under the rear seat** Space for the cabin outlet box: length × width × height, what's there now	45 L `CABIN_SEAT_BOX`	L W H	5	☐
E6	**Cab floor grommets** Existing grommets that could take the cabin feed from the chassis rail (R20)	–		5	☐
E8	**Cab back (C-pillar corner)** Flat face up the passenger corner of the cab back for the 32 mm conduit (R25), and the gap between the cab top and the platform rear member	–	Width Gap	2	☐
E9	**Chassis rail web** Height of the passenger rail web and the free length for the two conduits from the tub to the cab-tub gap (R16 / R17)	180 / –	Web H Free L	1	☐
E7	**Dash switch blanks** Free switch blanks on the dash (optional visor stow switch)	–		7	☐

9 · Equipment and vehicle details

ID	Measure	Model now	Your values (mm unless noted)	Stage	📷
Q1	**Fridge** Confirmed Dometic CFX3 45 (694 × 398 × 476): check the handle height and that the handles fold up for the straps	CFX3 45	Model ☐ L ☐ W ☐ H ☐ Lid ☐	3	☐
Q2	**Accessories fitted** Tow bar (ball weight), bull bar, side steps, snorkel, anything else	–	.....	-	☐
Q3	**Compliance plate** GVM, GCM, rear axle rating	GVM 3250	GVM ☐ GCM ☐ Rear axle ☐	-	☐
Q4	**Weighbridge (optional)** Front and rear axle weights, unladen, full fuel	–	Front ☐ Rear ☐	-	☐
Q5	**Tyres** Size and pressures	265/70R17	Size ☐ Pressures ☐	-	☐

Photo checklist

- [ ] Whole tub from the tailgate, from each rear corner and from above (lid open)
- [ ] Each arch: side on, with a tape along it; close-up of each arch well with a rule in it
- [ ] Front wall and both front corners; the floor ribs with a rule across them
- [ ] Factory outlet: on, and removed showing the hole and what's behind
- [ ] Every tie-down and protrusion with a rule beside it
- [ ] Under the lid: hinges, lock, struts, clamps, any frame members
- [ ] Each rear corner (both sides): square-on to the side wall from the tub centre, from the tailgate looking forward, and from above with the lid open, with a tape along the floor and up the tail-light housing
- [ ] Destroyer rail end-on (T-slot profile) with calipers
- [ ] Each of the 4 tent-bolt crossings from the side, with a rule from the bar to the lowest point
- [ ] Cab–tub gap from above and below; chassis rail beside the gap
- [ ] Roof: straight edge across with the gaps visible; the roof mounting points uncovered
- [ ] Side profile of the windscreen and bonnet with a level for reference
- [ ] Engine bay: start battery, fuse box, route to the chassis; under the rear seat

Notes and sketches